Project Title: AI Customer Support Coaching Assistant

Project Statement: Customer service representatives handle dozens of support interactions daily, often without access to real-time guidance, relevant knowledge resources, or immediate feedback on response quality. Training agents through post-call reviews is slow and reactive, failing to improve performance during live interactions where it matters most. Organizations need intelligent systems that coach agents in the moment — improving resolution quality, reducing escalations, and enhancing customer experience continuously.

The objective of this project is to develop an **AI Customer Support Coaching Assistant** — an intelligent platform that coaches customer service agents in real time during live text-based support interactions. The platform supports three interaction modes: **Simulator Mode** where a Customer Simulator Agent generates realistic customer messages turn by turn based on a defined scenario, **Manual Mode** where the agent pastes incoming customer messages, and **Replay Mode** where a pre-loaded support transcript is replayed message by message. After each exchange, a multi-agent pipeline analyzes customer intent and sentiment, retrieves relevant knowledge base content via RAG, generates response suggestions, evaluates agent tone and clarity, and monitors escalation risk — all updating in real time within a three-panel coaching console. Post-interaction, the system generates a structured performance report with sentiment journey, resolution quality score, and personalized coaching recommendations.

The platform transforms reactive agent training into proactive in-session coaching, improving first-interaction resolution rates and agent performance continuously.

Outcomes:

1. Develop a multi-agent real-time coaching pipeline that analyzes each conversation turn and delivers instant guidance to support agents.
2. Build a Customer Simulator Agent that generates realistic, scenario-based customer messages for self-contained simulation without real users.
3. Provide RAG-powered knowledge recommendations — surfacing relevant FAQs, support articles, and troubleshooting steps per conversation context.
4. Detect escalation risk continuously with reasoning and recommended intervention strategies.
5. Generate post-interaction performance reports with sentiment journey, resolution quality scoring, and personalized improvement recommendations.

Agents to be Implemented:

1. **Customer Simulator Agent** — Generates realistic customer messages turn by turn based on defined problem scenario and persona
2. **Intent & Sentiment Analysis Agent** — Identifies customer intent, emotional state, frustration level, and satisfaction trend per message
3. **Knowledge Recommendation Agent** — RAG-powered retrieval of relevant support articles, FAQs, and troubleshooting steps based on conversation context

4. **Coaching & Response Suggestion Agent** — Generates suggested agent responses, tone quality feedback, and communication improvement tips
5. **Escalation Risk Monitor Agent** — Continuously scores escalation likelihood with reasoning and recommended resolution strategies
6. **Post-Interaction Summary Agent** — Generates interaction summary, sentiment journey, resolution quality score, and coaching recommendations

Modules to be Implemented:

1. Session Configuration and Interaction Mode Selection Module
2. Support Knowledge Base Ingestion and RAG Pipeline Module
3. Customer Simulator Agent and Conversation Management Module
4. Multi-Agent Coaching Pipeline and Orchestration Layer
5. Live Support Console UI — Three-Panel Coaching Interface
6. Escalation Risk Detection and Alert Module
7. Post-Interaction Report and Performance Analytics Module

Milestones:

Milestone 1 (Week 1-2)

1. Study customer support workflows, coaching methodologies, RAG architecture, and multi-agent design patterns.
2. Design system architecture, agent responsibilities, orchestration flow, and data models.
3. Build session configuration module — user selects interaction mode (Simulator, Manual, Replay), defines product context and customer scenario.
4. Develop Support Knowledge Base ingestion — upload FAQs, support docs, and policy PDFs; implement chunking, embedding, and vector indexing via RAG pipeline.

Milestone 2 (Week 3-4)

1. Build Customer Simulator Agent — generates realistic, scenario-consistent customer messages turn by turn with configurable persona and emotional progression.
2. Develop Intent and Sentiment Analysis Agent — identifies customer intent, emotional state, frustration level, and satisfaction trend after each message.
3. Build Knowledge Recommendation Agent — retrieves contextually relevant support articles, FAQs, and troubleshooting steps from RAG knowledge base per conversation turn.
4. Test Simulator Agent realism and Knowledge Agent retrieval accuracy across multiple support scenarios.

Milestone 3 (Week 5-6)

1. Build Coaching and Response Suggestion Agent — generates suggested agent responses, evaluates tone and clarity, and provides communication improvement tips per turn.
2. Develop Escalation Risk Monitor Agent — continuously scores escalation likelihood with reasoning, triggers alert overlay when risk reaches high threshold.
3. Build Live Support Console UI — three-panel layout with conversation window, real-time coaching feed, and knowledge recommendation panel updating after each exchange.
4. Implement Manual Mode and Replay Mode — agent pastes messages or uploads transcript for message-by-message coaching replay.

Milestone 4 (Week 7-8)

1. Build Post-Interaction Summary Agent — generates interaction summary, customer sentiment journey timeline, resolution quality score, and personalized agent coaching recommendations.
2. Develop Performance Analytics Module — tracks trends across multiple sessions including common escalation triggers, knowledge gaps, and agent improvement indicators.
3. Conduct end-to-end testing across all three interaction modes, all agents, and post-interaction reporting.
4. Prepare technical documentation, project report, and final demonstration.

Evaluation Criteria:

1. Realism and scenario consistency of Customer Simulator Agent generated messages.
2. Accuracy of Intent and Sentiment Analysis Agent across varied customer emotional states and intents.
3. Relevance and retrieval quality of RAG-powered Knowledge Recommendation Agent suggestions.
4. Usefulness and actionability of real-time coaching, response suggestions, and escalation alerts.
5. Quality and clarity of post-interaction performance reports and sentiment journey visualization.
6. Completeness of implementation, testing, documentation, and final demonstration.